RELATIVE PSEUDO-COHERENCE TESTED AFTER A CLOSED IMMERSION INTO AFFINE SPACE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 16625-16663, immediately following the statement at 16609-16623.

Editorial note. Three cross-chapter pointers (More on Algebra, Derived Categories of Schemes) are recorded in assembly.external_references.

Editorial note. The statement's two conditions (some embedding / all embeddings) are kept verbatim in the statement span.

Allowed standard prerequisites (editorial).

- Ring-level criterion for relative pseudo-coherence. More on Algebra, lemma more-algebra-lemma-relatively-pseudo-coherent, cited in the proof. Supplies the algebraic counterpart of the equivalence, phrased for a complex of A-modules over R[x_1..x_n].
- Pseudo-coherence and pushforward along a closed immersion. Derived Categories of Schemes, lemma perfect-lemma-closed-push-pseudo-coherent, cited in the proof. Moves pseudo-coherence across a closed immersion, which is what makes the choice of embedding irrelevant.
- Affine comparison for pseudo-coherent objects. Derived Categories of Schemes, lemma perfect-lemma-pseudo-coherent-affine, cited in the proof. Identifies pseudo-coherence of the pushed-forward object with the module-level condition on global sections.

Source reference. Original reference 065H: focused-seeds/geometry/S9-morph-relatively-pseudo-coherent-embedding/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09VA: focused-seeds/geometry/S9-morph-relatively-pseudo-coherent-embedding/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08E7: focused-seeds/geometry/S9-morph-relatively-pseudo-coherent-embedding/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $X \to S$ be a finite type morphism of affine schemes. Let E be an object of $D(\mathcal{O}_X)$. Let $m \in \mathbf{Z}$. The following are equivalent*

(1) *for some closed immersion $i : X \to \mathbf{A}^n_S$ the object Ri_*E of $D(\mathcal{O}_{\mathbf{A}^n_S})$ is m-pseudo-coherent, and*

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(2) *for all closed immersions $i : X \to \mathbf{A}^n_S$ the object Ri_*E of $D(\mathcal{O}_{\mathbf{A}^n_S})$ is m-pseudo-coherent.*

Proof. Say $S = \mathrm{Spec}(R)$ and $X = \mathrm{Spec}(A)$. Let i correspond to the surjection $\alpha : R[x_1, \ldots, x_n] \to A$ and let $X \to \mathbf{A}^m_S$ correspond to $\beta : R[y_1, \ldots, y_m] \to A$. Choose $f_j \in R[x_1, \ldots, x_n]$ with $\alpha(f_j) = \beta(y_j)$ and $g_i \in R[y_1, \ldots, y_m]$ with $\beta(g_i) = \alpha(x_i)$. Then we get a commutative diagram

$$\begin{array}{ccc} R[x_1, \ldots, x_n, y_1, \ldots, y_m] & \xrightarrow[y_j \mapsto f_j]{} & R[x_1, \ldots, x_n] \\ \downarrow{\scriptstyle x_i \mapsto g_i} & & \downarrow \\ R[y_1, \ldots, y_m] & \longrightarrow & A \end{array}$$

corresponding to the commutative diagram of closed immersions

$$\begin{array}{ccc} \mathbf{A}^{n+m}_S & \longleftarrow & \mathbf{A}^n_S \\ \uparrow & & \uparrow \\ \mathbf{A}^m_S & \longleftarrow & X \end{array}$$

Thus it suffices to show that under a closed immersion

$$f : \mathbf{A}^m_S \to \mathbf{A}^{n+m}_S$$

an object E of $D(\mathcal{O}_{\mathbf{A}^m_S})$ is m-pseudo-coherent if and only if Rf_*E is m-pseudo-coherent. This follows from Derived Categories of Schemes, Lemma 09VA and the fact that $f_*\mathcal{O}_{\mathbf{A}^m_S}$ is a pseudo-coherent $\mathcal{O}_{\mathbf{A}^{n+m}_S}$-module. The pseudo-coherence of $f_*\mathcal{O}_{\mathbf{A}^m_S}$ is straightforward to prove directly, but it also follows from Derived Categories of Schemes, Lemma 08E7 and More on Algebra, Lemma 065H. □